

Lesson 6

Bandpass digital transmission

1 Basic problems

This first part includes problems extracted from the bibliography, aimed at practicing basic calculations that must be mastered.

Problem 6.1

[Sklar2001] Calculate the expected value of the number of error bits during a day for the coherent BPSK receiver described below, under continuous operation. The data rate is 5000 bits/s . The input digital signals are $s_1(t) = A \cdot \cos(\omega_p t)$ and $s_2(t) = -A \cdot \cos(\omega_p t)$, where $A = 1 \text{ mV}$, and the unilateral noise power spectral density is $N_0 = 10^{-11} \text{ W/Hz}$.

RESULTS FOR PROBLEM

2338 bits

Problem 6.2

[Sklar2001] A coherent BPSK system operating continuously produces errors at an average rate of 100 errors per day. The data rate is 1000 bits/s . The unilateral noise power spectral density is $N_0 = 10^{-10} \text{ W/Hz}$.

- If the system is ergodic, which is the average error probability?
- If the average received power is adjusted to 10^{-6} W , would this value be enough to keep the error probability calculated in a)?

RESULTS FOR PROBLEM

- $1.16 \cdot 10^{-6}$
- No

Problem 6.3

[Haykin2001] The signal component of a coherent PSK system is defined by the expression

$$s(t) = A_c k \sin(\omega_p t) \pm A_c \sqrt{1 - k^2} \cos(\omega_p t)$$

where $0 \leq t < T_b$, and the plus sign corresponds to the 1 symbol, and the minus sign corresponds to the 0 one. The first term on the right hand side of the equation represents a carrier component, included to improve the synchronization between transmitter and receiver. Solve this:

- a) Plot the constellation of the signals described; what can be said about this diagram?
- b) Show that, in presence of zero-mean additive white Gaussian noise with power spectral density $N_0/2$, the average error probability is

$$P_e = Q \left(\sqrt{\frac{2E_b}{N_0}(1 - k^2)} \right)$$

with $E_b = \frac{1}{2} A_c^2 T_b$

- c) Assume that 10% of the transmitted power is located in the carrier component. Determine the value of E_b/N_0 required to obtain an error probability of 10^{-4} .
- d) Compare this E_b/N_0 value with the one required in a conventional PSK system with the same error probability.

RESULTS FOR PROBLEM

- a) PSK constellation
- b) Demonstration
- c) $\frac{E_b}{N_0} = 8.02$
- d) $\frac{E_b}{N_0} = 7.22$

Problem 6.4

[Haykin2001] We want to compare two data transmission bandpass systems. One of them employs 16-PSK, the other, 16-QAM. Both systems have to provide an average symbol error probability of 10^{-3} . Compare the signal-to-noise requirements of said systems.

RESULTS FOR PROBLEM

$$\Delta \left(\frac{E_s}{N_0} \right) = 3.68dB$$

Problem 6.5

[Sklar2001] If the performance criterion of a system is the bit error probability, which one of the following modulation schemes would be chosen to operate in an AWGN channel? Show the calculations.

- a) Non-coherent binary orthogonal FSK with $E_b/N_0 = 13dB$.
- b) Coherent binary PSK with $E_b/N_0 = 8dB$.

RESULTS FOR PROBLEM

Non-coherent binary orthogonal FSK

Problem 6.6

[Haykin2001] A system transmits binary data at a rate of $2.5 \cdot 10^6$ bps. During transmission, zero-mean additive white Gaussian noise with power spectral density $N_0/2 = 10^{-20}$ W/Hz is added to the signal. In the absence of noise, the amplitude of the received sinusoidal signal for digit 1 or 0 is $1 \mu V$. Determine the average symbol error probability for the following system configurations:

- Coherent binary FSK.
- Coherent MSK.
- Non-coherent binary FSK.

RESULTS FOR PROBLEM

- $9,67 \cdot 10^{-4}$
- $5,41 \cdot 10^{-6}$
- $3,36 \cdot 10^{-3}$

Problem 6.7

(May 2016 exam) A digital communications system uses an FSK modulation scheme to transmit two orthogonal and equiprobable signals, $s_1(t)$ and $s_2(t)$, at a rate of $2.5 \cdot 10^6$ bps. The system is characterized by $E_b/N_0 = 12$ dB.

$$s_i(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(\omega_i t), & 0 \leq t < T_b \\ 0, & \text{otherwise} \end{cases}, \quad i = 1, 2.$$

- Calculate the average error probability for FSK modulation schemes with coherent and non-coherent receivers.
- Which of the previous schemes is more suitable for a channel modeled with AWGN noise? Justify your answer. For the rest of the exercise, consider only the scheme selected in this section.
- Calculate the minimum bandwidth of the scheme selected in part b).
- For that scheme, determine an orthonormal basis that allows representing $s_1(t)$ and $s_2(t)$. Plot the constellation in signal space indicating the coordinates of the signal vectors.
- If you compare the previous modulation with a coherent BPSK with the same P_e , which one will require a higher E_b/N_0 ratio? Compute the increase in E_b/N_0 expressed in dB.

RESULTS FOR PROBLEM

- $P_e^{coh} = 4.81 \cdot 10^{-5}$, $P_e^{no-coh} = 1.81 \cdot 10^{-4}$
- FSK with coherent receiver
- $B_{min} = 6.25$ MHz

- d) Basis: normalized $s_1(t)$ and $s_2(t)$.
- e) FSK. E_b/N_0 increase of 3 dB.

References

- [Haykin2001] Simon Haykin. Communication Systems, 4th Ed. John Wiley and Sons, 2001.
- [Sklar2001] Bernard Sklar. Digital Communications, 2nd Ed. Prentice Hall, 2001.